Mock R2, the written half: can you check a vendor's card model on paper?

Thirty minutes, alone, on paper. No laptop, no phone, no AI and no notes. Write each letter in the answer box on the last page, and show your working under each Part C item.

What has the vendor sent Rohan?

A vendor offers Rohan Desai a scoring model for card defaults and shares its test results on a card book of its own. **Every number on this sheet is invented**, and none of it comes from your group's data.

The vendor's book	
Accounts	10,000
Default next month	2,000 accounts, 20 percent
Held out as a test set	2,000 accounts, of whom 400 default
A missed default (not called, then defaults)	costs 5 units
A wasted call (called, then pays)	costs 1 unit

The **error bill** of a list is 1 unit for every wasted call plus 5 units for every missed default. A **threshold** is the chance of default above which an account is called.

Exhibit 1. The vendor's model at three thresholds, on the 2,000 test accounts.

Threshold	Accounts called	Defaulters among them	Accuracy
0.50	220	160	85.0 percent
0.17	900	330	68.0 percent
0.05	1,800	395	29.5 percent

Part A: what does each number tell Rohan?

One correct option each, about two minutes each.

1. On the test set, what does calling nobody score?

a) 20 percent accuracy, a bill of 2,000 units

b) 80 percent accuracy, a bill of 2,000 units

c) 80 percent accuracy, a bill of 400 units

d) 80 percent accuracy, a bill of 0 units

2. Above what chance of default does a call pay for itself?

a) 0.50

b) 0.20

c) 0.17

d) 0.83

3. The list is built on the 1st, before this month's statement. Which column may it use?

a) Last month's statement balance

b) Payments missed by the month's end

c) Whether collections called this month

d) The amount paid after the due date

4. The model's top 200 test accounts hold 110 defaulters. How many times the defaulters of a random 200 is that?

a) 0.55 times

b) 1.38 times

c) 5.50 times

d) 2.75 times

5. Boosting scores AUC 0.95 on its training rows and 0.74 on validation; logistic regression scores 0.77 and 0.76. Which call fits?

a) Ship logistic: boosting memorised

b) Ship boosting: 0.95 beats 0.77

c) Ship neither: both AUCs are below 0.80

d) Ship boosting: its gap will close in use

6. Rohan wants one number to pay his callers by. Which keeps them working on what the list is for?

a) Calls completed per caller per day

b) Share of the list called by the due date

c) Defaulters reached before the due date

d) Average call length, in minutes

Part B: true or false, and why?

Pick the option whose verdict and reason are both right, about two and a half minutes each.

7. Fitting the scaler on all 10,000 accounts before five-fold cross-validation gives an honest estimate. True or false?

a) True, because a scaler never sees the labels

b) False, because scaling changes who defaults

c) True, because every fold then shares the same scale

d) False, because the scaler saw the held-out folds

8. Since the model's accuracy is highest at 0.50 in Exhibit 1, should Rohan use 0.50?

a) True, because accuracy counts both mistakes

b) False, because 0.17 costs 340 units less

c) True, because 0.50 calls the fewest accounts

d) False, because accuracy is highest at 0.17

Part C: what does the arithmetic say?

Write the number with its unit, and the working that gets you there; about four minutes each.

9. Using Exhibit 1, what are the error bills of the 0.05 list and of calling every test account, and which costs Rohan less, by how much?

10. Rohan's team can make only 300 calls. The model's top 300 hold 150 defaulters; the rule "sort by months late" puts 138 defaulters in its top 300. What is each list's error bill on the test set, and how many more defaulters does the model reach?

11. Over the last six months the vendor's book defaulted at 18, 19, 21, 20, 22 and 21 percent. For next month the model forecasts 19 percent and "same as last month" forecasts 21 percent. The month comes in at 22 percent. How many accounts of the 10,000 does each forecast miss by, and which forecast was better?

Where do your answers go?

Name: _____ Group: _____

Item	Your letter
1	
2	
3	
4	
5	
6	
7	
8	

Item	Your number, with its unit
9	
10	
11	